Evidence for the Efficacy of Conflict-reducing Practices in Undergraduate Evolution Education in a Randomized Controlled Study

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ABSTRACT

Conflict-reducing practices during evolution instruction have been recommended to increase students' perceived compatibility between evolution and religion, increase evolution acceptance, and decrease stereotypes about religious students in science. However, the efficacy of these practices has not been demonstrated in a randomized controlled design making it uncertain whether they are causing the effects reported in less controlled studies. Further, we do not know the extent to which the religious identities of instructors may impact their effectiveness. In this study, we randomly assigned 2623 undergraduate students in 19 biology courses across different states to receive an evolution video with 1) no conflict-reducing practices, 2) conflict-reducing practices implemented by a non-religious instructor, or 3) conflict-reducing practices implemented by a Christian instructor. We found that the evolution videos with conflict-reducing practices led to decreased conflict, increased compatibility, and increased acceptance of human evolution compared with the video without conflict-reducing practices. Further, the Christian and non-religious instructor conditions were equally effective at improving all student outcomes, except the non-religious instructor was more effective for increasing perceived compatibility between religion and evolution among atheist students. These results illustrate that conflict-reducing practices, implemented by either Christian or non-religious instructors, can be effective in a controlled study.

INTRODUCTION AND BACKGROUND

Evolution is a foundational topic in biology (AAAS, 2011; Brownell *et al.*, 2014), yet human evolution has been one of the most controversial topics in American society for more than 150 years (Numbers, 2006). Evolution education experts have defined *evolution acceptance* as "agreeing that evolution is valid and the best explanation from science for the unity and diversity of life on Earth, which includes speciation, the common ancestry of life, and that humans evolved from non-human ancestors" (Barnes *et al.*, 2024). Despite an overwhelmingly strong consensus among scientists about the scientific validity of evolution, around half of the American public do not agree that humans have evolved from non-human species (Brenan, 2019; Pew Research Center, 2019) and around one-third of undergraduate biology students sampled from across the United States do not think all life shares a common ancestor, including human evolution (Barnes, Dunlop *et al.*, 2020).

This rejection of core components of evolution among undergraduate biology students is important for several reasons. First, evolution is considered to be one of the core concepts or big ideas of biology (AAAS, 2011; NGSS Lead States, 2013)

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and has been argued to be a conceptual thread that underlies all of biology and should be integrated throughout the curriculum (Dobzhansky, 1973). If students reject major components of evolution, it is unlikely that they will incorporate evolutionary thinking into their conceptual framework for biology (Smith, 2010; Smith and Siegel, 2016). Further, if these students go into allied health careers or biomedical research careers, they may be resistant to using evolutionary medicine as a lens to examine human health and disease, including the consideration of evolutionary similarities and differences among species for drug development (Uhl and Warner, 2015; Grunspan *et al.*, 2018; Natterson-Horowitz *et al.*, 2023).

Second, acceptance of strongly supported claims such as human evolution can be an important indicator of the extent to which students can appropriately evaluate a scientific claim, also known as their scientific literacy (Smith, 2010; Sickel and Friedrichsen, 2013; Nadelson and Hardy, 2015; Smith and Siegel, 2016; Kampourakis, 2022). The evidence for the history of the human lineage evolving from non-human apes is supported by multiple converging lines of evidence, a hallmark of a strongly supported scientific claim (Goodman and Sterner, 2010; Winder and Winder, 2014; Andrews, 2019; Almécija *et al.*, 2021). An especially important part of scientific literacy is being able to appropriately evaluate scientific claims even when they contradict or conflict with previously held beliefs (Nadelson and Sinatra, 2009; Heddy and Nadelson, 2012; Sinatra *et al.*, 2014). Thus, helping students understand *and* accept evolution have been deemed important if we aim to foster scientific literacy among students (Lombrozo *et al.*, 2008; Kampourakis, 2022; Pessoa *et al.*, 2024).

Third, evolution acceptance also shapes how individuals engage with knowledge at the intersection of science and society (Kauffman *et al.*, 2019; Kampourakis, 2022). Skepticism about evolution has a history of driving public support for incorporating special creationism and intelligent design into K-12 education as alternatives to evolution (Berkman *et al.*, 2008; Forrest, 2010) or even removing any mention of evolution from the curriculum (Wiles, 2005), underlying how evolution acceptance can have downstream effects on how we educate the next generation. Evolution can help explain modern medical phenomena such as antibiotic resistance due to the overuse of antibiotics (Jenkins and Antolin, 2011; Antonovics, 2016), or the concept of herd immunity and the importance of vaccinations during pandemics (Hanley, 2011; Betsch *et al.*, 2017), making it relevant to health care policy decisions typically enacted by non-scientists. Additionally, human evolution acceptance could play a role in addressing biological racism by emphasizing that all humans share recent common ancestry, illustrated by the lack of genetic diversity within our species, helping to debunk the notion of distinct biological groups (Donovan *et al.*, 2021; Guevara *et al.*, 2023). In sum, these reasons highlight the need for educators to develop strategies to promote evolution acceptance, and not just understanding, among their students.

Perceived Conflict Between Evolution and Religion Influences Evolution Acceptance

Research illustrates that rejection of evolution is a multifaceted and complex phenomenon with many related factors (e.g., understanding of evolution [Weisberg *et al.*, 2018], nature of science understanding [Lombrozo *et al.*, 2008; Dunk *et al.*, 2017], and family attitudes toward evolution [Barnes, Evans *et al.*, 2017]). In the United States, however, variables related to religion are the strongest predictive factors. Both an individual's particular religious affiliation and their religiosity, or the extent to which they identify as religious, highly correlate with their evolution acceptance (Glaze *et al.*, 2015; Dunk *et al.*, 2017; Jensen *et al.*, 2019). Our research groups have reported that undergraduate biology students who are more religious tend to accept evolution less (Aini *et al.*, 2024), and Christian and Muslim students tend to have lower evolution acceptance than students with other religious identities (Barnes *et al.*, 2019; 2021). More than 65% of undergraduate biology students from large samples across the nation identify as religious, and over 50% of undergraduate biology students specifically identify as Christian (Barnes, Supriya *et al.*, 2020; Barnes, Roberts *et al.*, 2021), so a majority of undergraduate biology students have a religious identity commonly associated with evolution rejection.

However, in recent years, researchers have shown that the *strongest* predictor of religious students' evolution acceptance is their *perceived conflict between evolution and religion*, as those perceptions of conflict help explain the relationship between students' religious identities and their acceptance of evolution (Barnes, Supriya *et al.*, 2021). Some students' religious beliefs are incompatible with the principles of evolution. For example, literal interpretations of the creation stories found within the Christian Bible directly contradict the evolutionary tenet that all life on Earth shares a common ancestor (Baker, 2012). However, there are many examples of highly religious Christian individuals who still accept evolution because they *do not perceive conflict* between evolution and their religion, and there are ways of reconciling a belief in a higher power with an acceptance of evolution (Miller, 1999; Scott, 2005; Collins, 2006; The Clergy Letter Project, 2023). For example, scholars have proposed that religion and science can be seen as addressing different questions and thus do not *have to* conflict with one another (Barbour, 1990; Gould, 1999), especially if individuals adopt metaphorical interpretations of religious texts (Winslow *et al.*, 2011). Despite that room for compatibility, in a 2020 study surveying 1898 undergraduate biology students nationwide, our groups found that over half of the sample thought someone must be an atheist to accept evolution (Barnes, Dunlop *et al.*, 2020), and this perception predicted rejection of evolution among religious students (Barnes, Dunlop *et al.*, 2020).

A Potential Solution: Conflict-Reducing Practices

Interview studies of undergraduate biology instructors teaching evolution have documented a hesitation about addressing religion when discussing evolution (Barnes and Brownell, 2016; Edwards *et al.*, *n.d.*). Prior studies have shown that this avoidance of the topic can be because instructors worry that addressing religion in a science class would be inappropriate or irrelevant to their classrooms and, thus, opt to remain "neutral" toward the topic by not bringing it up at all (Barnes and Brownell, 2016; Stahi-Hitin and Yarden, 2022). However, some Christian students have reported that they feel as though an instructor's decision to avoid mentioning religion during discussions of evolution does not indicate neutrality

(Barnes, Truong *et al.*, 2017). Due to general perceptions that scientists are not religious (Rios *et al.*, 2015; Mackey *et al.*, 2023), Christian students can assume evolution instructors not discussing religion at all may be a sign that they think there is a conflict between religion and evolution and are not accepting of students' religious beliefs (Barnes, Truong *et al.*, 2017). Further, both students and instructors have reported instances in which biology instructors have made negative comments or jokes about religion and/or religious individuals during class (Barnes and Brownell, 2016; Barnes, Truong *et al.*, 2017), which we call *religion negativity*. This can adversely influence students' acceptance of evolution (Aini *et al.*, 2022), offend religious students (Cooper *et al.*, 2020), and convey an anti-religion message in science courses, perpetuating a narrative of conflict between evolution and religion that may subvert the possibility of students being open to the idea that religion and evolution could be compatible.

Conflict-reducing practices can be used to help instructors discuss the relationship between evolution and religion in a way that improves the outcomes of religious students in evolution education (Barnes and Brownell, 2017; Truong *et al.*, 2018; Lindsay *et al.*, 2019; Ferguson and Jensen, 2021; Aini, Supriya *et al.*, 2024). Importantly, these practices do not advocate for teaching any particular religious belief or giving credibility to anti-evolution stances or intelligent design, but rather involve acknowledging that while there is conflict between certain religious beliefs and evolution, it is possible for one to believe in a higher power and also accept evolution. The conflict-reducing practices that have been proposed as potentially beneficial for improving students' evolution education outcomes, such as their perceived conflict and compatibility between religion and evolution and their evolution acceptance, are as follows: 1) acknowledging that students have *autonomy* over their religious beliefs and acceptance of evolution, 2) highlighting the *spectrum of religious beliefs* and how the majority of beliefs are not necessarily in conflict with evolution, 3) presenting the *bounded nature of science*, specifically that science can answer certain questions and religion/faith may answer different questions, 4) presenting examples of religious leaders and scientists who accept evolution as *role models* for religious students, and 5) *avoiding religion negativity* (Barnes and Brownell, 2017). There is a growing body of evidence across multiple contexts that supports the efficacy of these practices, including evidence from studies examining one practice in isolation and studies documenting the impact of a collection of these practices on students (Winslow *et al.*, 2011; Southerland and Scharmann, 2013; Manwaring *et al.*, 2015; Barnes, Elser *et al.*, 2017; Truong *et al.*, 2018; Bertka *et al.*, 2019; Lindsay *et al.*, 2019; Barnes, Werner *et al.*, 2020; Green and Delgado, 2021; Bowen *et al.*, 2022). While most of these studies have been done in single classrooms or at one institution, a quasiexperimental national study of students in 55 biology courses at universities in multiple states across the United States has found that biology instructors highlighting religious scientist role models and acknowledging student autonomy can reduce student perceived conflict and increase student evolution acceptance (Aini, Supriya *et al.*, 2024). Further, conflict-reducing practices have been shown to be successful across a variety of formats and settings, including an online course (Bowen *et al.*, 2022), non-majors biology courses (Green and Delgado, 2021; Bowen *et al.*, 2022), and at public community colleges (Green and Delgado, 2021), public 4-year universities (Truong *et al.*, 2018), and private Christian 4-year colleges with highly religious students (Lindsay *et al.*, 2019; Barnes, Werner *et al.*, 2020; Ferguson and Jensen, 2021). Further, while the majority of studies have been done with undergraduate biology students, there seems to be efficacy of the conflict-reducing practices for theology students (Tolman *et al.*, 2020), public aquarium patrons (Tolman *et al.*, 2021), and secondary science teachers and youth religious educators (Kaloi *et al.*, 2022).

Conflict-reducing practices have been called *reconciliation* practices when they are implemented by religious instructors for religious students (Lindsay *et al.*, 2019) and *religious cultural competence in evolution education (ReCCEE)* practices when they are implemented by non-religious instructors to help communicate with religious students (Barnes and Brownell, 2017). However, we have recently adopted the terminology of *conflict-reducing practices* to more clearly illustrate two key points: 1) it is the reduction of perceived conflict between religion and evolution that underlies the efficacy of these practices (Aini, Supriya *et al.*, 2024), and 2) there is potential broad applicability of these practices for reducing perceived conflict between religion and evolution with all instructor-student identity pairs (e.g., religious instructors with religious students, non-religious instructors with religious students, and even religious or non-religious instructors with non-religious students as we discuss below).

Why Focus on Both Non-Religious and Religious Students?

While conflict-reducing practices were originally proposed for reducing perceived conflict to increase religious student evolution acceptance, they can be potentially useful for diminishing perceived conflict to increase the *religious cultural competence* of *non-religious* students when they are communicating with religious individuals. *Cultural competence* is the ability of one group to communicate effectively with individuals of another group that differs in some notable way (Betancourt *et al.*, 2003). For example, cultural differences between physicians and their patients can lead to ineffective communication and ultimately healthcare. However, if physicians become more aware of sociocultural differences and how these differences may affect patient healthcare, and thus become more culturally competent, then the patient experience and healthcare are likely to improve. The same idea has been applied to instructors teaching students with different sociocultural backgrounds (Gay, 2018), typically with a focus on language and ethnicity.

Our past research illustrates that religious undergraduate biology students can have negative experiences in the context of academic biology based on their religious identity. While some of this stems from what instructors may say or do, Christian and Muslim students have reported that their non-religious *peers* can make negative comments about religion and religious beliefs during evolution instruction, that non-religious students can assume their religious peers are not capable of accepting evolution, and that non-religious students may not even think that one can be both a

scientist and be religious (Barnes, Truong *et al.*, 2017; Barnes, Maas *et al.*, 2021; Edwards *et al.*, 2024; Google *et al.*, 2023; Edwards *et al.*, n.d.). These experiences support the assertion that there could be a benefit in having non-religious students become more culturally competent about religion in biology classes, particularly in class settings like active learning where there are greater levels of student interaction (Henning *et al.*, 2019; Edwards *et al.*, 2024). We have some indication that conflict-reducing practices can impact non-religious students. For example, non-religious students reported that before evolution instruction they perceived that religious people always reject evolution but they changed this perception when their evolution instructor used conflict-reducing practices (Barnes, Elser *et al.*, 2017). Further, when atheist students assumed a conflict between evolution and religion, we have found that they were not recognizing the bounded nature of science and concluded that science disproves the existence of a God/god(s) (Coscia *et al., n.d.*). Teaching the bounded nature of science, which clarifies that science is limited to natural phenomena and does not address metaphysical claims, is considered a conflict-reducing practice (Southerland and Scharmann, 2013; Barnes and Brownell, 2017). Thus, beyond increasing evolution acceptance among religious students, implementing conflict-reducing practices can help encourage culturally competent communication between non-religious and religious individuals.

Need For Randomized Controlled Studies of Conflict-Reducing Practices With Religious and Non-Religious Students

Even though there is growing evidence for the impact of conflict-reducing practices, there is still much that remains unknown about their efficacy on different groups of students. For example, there have been multiple qualitative studies (Truong *et al.*, 2018; Bertka *et al.*, 2019; Green and Delgado, 2021), quantitative studies with pre-post designs (Manwaring *et al.*, 2015; Barnes, Elser *et al.*, 2017; Lindsay *et al.*, 2019; Barnes, Werner *et al.*, 2020; Green and Delgado, 2021; Bowen *et al.*, 2022), and most recently a large quasi experimental study using the natural variation of these practices among instructors (Aini, Supriya *et al.*, 2024). However, few studies have examined the differential impact of these practices on students with different religious beliefs. For instance, although we have self-reports that conflict-reducing practices reduced non-religious student perceived conflict between religion and evolution, we know of no controlled studies that have examined this impact, even though non-religious individuals tend to have higher perceptions of conflict between science and religion than Christians (Coscia *et al., n.d.*; Leicht *et al.*, 2022).

Further, while the strength of quasi experimental studies is that they occur in real classrooms, the disadvantage of that research design is that it is nearly impossible to control for most aspects of the intervention. In quasi experimental studies using pre-post design, we cannot disaggregate other aspects of instruction such as how the content is taught, instructor immediacy (Witt *et al.*, 2004; Creasey *et al.*, 2009), instructor talk (Seidel *et al.*, 2015; Ovid *et al.*, 2021) or any other instructor effects that researchers have found can influence student experiences and perceptions (Erdle *et al.*, 1985; Fleck and Richmond, 2024). Given the accumulated evidence for the effectiveness of conflict-reducing practices, it is imperative that we move beyond studying *whether* conflict-reducing practices are effective, to also examine for *whom* they are effective and what are ways to *maximize impact* on students from different religious and non-religious backgrounds; the only way that we can begin to make strategic decisions about this is to conduct highly controlled studies where only a small number of variables are altered, including randomized controlled studies.

Theoretical Frameworks: Social Identity Theory, Intergroup Contact Theory, and Cultural Competence

We hypothesize that the efficacy of conflict-reducing practices in evolution education for non-religious students may differ from those of religious students and may depend on the relationship between their identities and the religious and non-religious identities of the instructor delivering the practices. These hypotheses are based on the theoretical frameworks of social identity and intergroup contact theory.

Social identity theory posits that individuals form their identities based on their memberships in social groups and that their beliefs and behaviors are shaped by their self-categorization within those groups (Tajfel and Turner, 2004). For instance, individuals perceive social *identity alignment* when they share characteristics with others, which leads to a positive bias toward those others (Brewer, 1999; Steele, 2011) and in turn affects the extent to which they trust and are influenced by them. This could mean that there will be larger effects on Christian student evolution education outcomes when they perceive a Christian instructor using conflict-reducing practices. It could also mean that there will be larger effects on non-religious students when they perceive a non-religious instructor using conflict-reducing practices. However, when individuals perceive *identity misalignment* (mismatched or incompatible identities), it may lead to a negative outgroup bias in which one is less likely to trust and be influenced by the other person (Steele *et al.*, 2002; Riek *et al.*, 2006). Thus, identity misalignment between instructors and students may weaken the impact of conflict-reducing practices.

Although identity misalignment may weaken impacts of conflict-reducing practices there may also be benefits if the interactions are positive. *Intergroup contact theory* proposes that positive interactions between members of *different* social identity groups with shared goals can reduce perceived conflict between members of those groups (Riek *et al.*, 2006; Ülger *et al.*, 2018). In the context of evolution instruction, most instructors do not disclose their religious or non-religious identities in class (Edwards *et al., 2025*), but by concealing them, instructors are decreasing opportunities for students to realize that they are experiencing intergroup contact, thus decreasing opportunities to reduce perceived tensions between science and religion communities. While social identity theory predicts positive outcomes based on *shared* identity and increased trust, intergroup contact theory would predict positive outcomes when those with *different* identities have *positive* interactions with one another. For Christian students viewing conflict-reducing practices from non-religious instructors, this could mean that negative outgroup biases against non-religious individuals will be decreased, reducing

perceived conflict, and increasing acceptance of evolution. For non-religious students viewing conflict-reducing practices from Christian instructors, this could mean a reduction of perceived bias against religious individuals in science and a decrease in perceived conflict between religion and evolution. We know of no controlled studies that have investigated the relative impact of using conflict-reducing practices when students have a different or the same religious or non-religious identity as the instructor delivering them.

It is important to note that for intergroup contact to have beneficial outcomes, the interaction has to be positive and thus culturally competent. As we previously mentioned, we have used this frame of cultural competence to increase the probability of having positive intergroup contact during evolution instruction. In college biology classrooms, religious cultural competence is most often relevant to non-religious instructors teaching Christian students since the majority of undergraduate biology instructors are not religious (Ecklund and Scheitle, 2007), and this is in contrast to the majority of United States undergraduate biology students who are predominantly Christian (Barnes, Supriya *et al.*, 2020; Barnes, Roberts *et al.*, 2021). This means that in the context of evolution instruction, the people who are in positions of power in the classroom (i.e., the instructors) tend to be non-religious and may not understand the sources of their students' rejection of evolution, presenting the potential for negative biases and interactions between students and instructors. We posit that non-religious instructors can use conflict-reducing practices to be culturally competent for their Christian students to increase positive intergroup contact between non-religious and religious cultural groups in science. Further, we posit that non-religious instructors, by using conflict-reducing practices, may increase their non-religious students' own cultural competence by reducing their perceived conflict between religion and evolution.

Current study

To address gaps in the literature and explore how instructor and student religious identity may influence the efficacy of conflict-reducing practices, we set out to answer the following research questions:

RQ1: Do conflict-reducing practices in evolution instruction lead to better evolution education outcomes compared with evolution instruction without conflict-reducing practices?

RQ2. Does an instructor revealing a non-religious or Christian identity impact the effectiveness of conflict-reducing practices for students of varying religious identities?

MATERIALS AND METHODS

Study design

Students in undergraduate introductory biology courses completed a homework assignment that included taking a presurvey, watching a video, and immediately completing a postsurvey. Students were told the assignment would prepare them for their upcoming evolution instruction in their courses. Their religious affiliation, evolution acceptance, evolution understanding, level of perceived conflict between evolution and religion, and level of perceived compatibility between evolution and religion were measured on the presurvey before they watched the video.

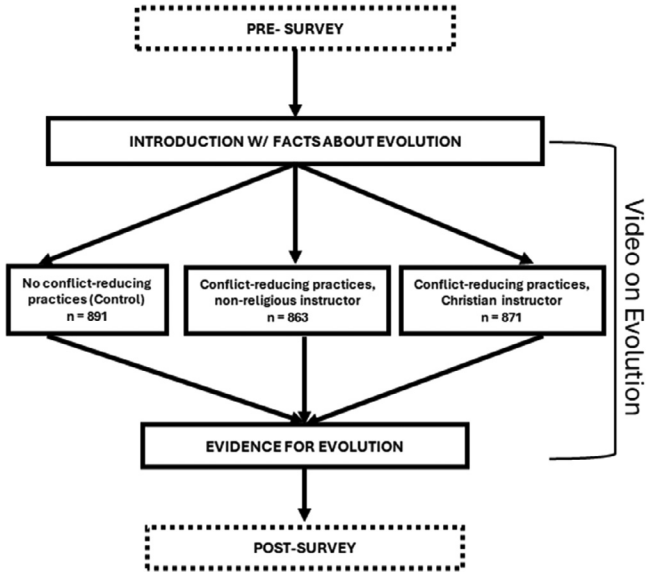


FIGURE 1. Illustration of study design. Students were surveyed before and after receiving a video that they were told would help prepare them for future evolution instruction. Students were randomized into three groups. All students received the same introduction to facts about evolution. The control group received no conflict-reducing practices, a second group received conflict-reducing practices with a non-religious instructor, and a third group received conflict-reducing practices with a Christian instructor. All students were then presented with a conclusion that included different lines of evidence for evolution.

Students were then blindly and randomly assigned to one of three video conditions to systematically test the impact of conflict-reducing practices and the instructor's religious identity on students' evolution education outcomes (Figure 1). All three video conditions used the same actor instructor to control for instructor effect, such as the instructor's other visible identities (in this case, a White woman), their personality, and how they taught. All students received the same introduction to evolution in the video in which the instructor defined what evolution is, what evolutionary biology helps explain about life, what evolutionary biologists study, and that evolution is considered the underlying thread of all of biology. Students were also presented with different sets of evidence for evolution including the fossil record, homologies, biogeography, and direct observations of speciation. As a control condition, one group watched the video of the instructor teaching about evolution without using conflict-reducing practices and without mentioning their religious identity (*Condition 1)*. This video was ~10 min. Two intervention groups watched the video of the same instructor teaching evolution using conflict-reducing practices (*Conditions 2 and 3)*. These videos were ~26 min. In these two intervention conditions, the actor instructor shared a personal narrative about grappling with the perceived conflict between evolution and religion during college—describing this struggle from the perspective of a non-religious person (*Condition 2*) and from the

perspective of a Christian (*Condition 3*) before explaining how they now teach evolution while highlighting potential compatibility between religion and evolution for their students' sake. We specifically chose to have the instructor reveal as non-religious rather than atheist or agnostic because we felt as though that would broaden the similar representation of this instructor to students who identified as atheist and agnostic. We used a Christian instructor because ~88% of religious students in biology classes are Christian (Barnes, Supriya *et al.*, 2021), which would increase the chances that these students experience identity alignment. We decided not to include a condition in which the instructor used conflict-reducing practices without revealing any religious identity because the absence of explicit cues in online instruction can lead students to form varied interpretations of the instructor's identity (Baker, 2012), which would increase noise between our conditions by promoting varied levels of perceived identity alignment among students. We were also limited by our sample size in testing three conditions, so we did not have adequate sample size to test conditions of non-Christian religious identities. To check whether randomization was achieved, we conducted group comparison analyses of demographic information between three conditions using chi-square (Supplemental Table S1) and found no significant associations in demographic distributions among the groups ($p > 0.05$). Additionally, a one-way ANOVA was also performed to compare the means between groups for all pretest variables (Supplemental Table S2), revealing no significant differences in pretest scores across the three conditions.

Immediately after watching their assigned video, students' evolution acceptance, evolution understanding, perceived conflict, and perceived compatibility between evolution and religion were measured again using a postsurvey.

All the items analyzed can be found in Supplemental Material S1.

Measurement

Perceived Conflict and Compatibility Between Evolution and Religion. We hypothesized that conflict-reducing practices would lead to improved perceptions of compatibility and conflict between religion and evolution. To assess students' perception of conflict and compatibility between evolution and religion, we used a previously developed measure to examine the extent to which someone agrees there is a conflict between evolution and religion as well as to what extent they agree that there is compatibility (Coscia *et al., n.d.*). Rather than measuring conflict and compatibility based on students' specific religious beliefs, this survey refers to conflict and compatibility between religion and evolution beyond one's own personal religious beliefs so any student, regardless of whether they are religious or not, can take it. This instrument consists of three items to measure perceived conflict between evolution and religion (example item: "*Scientists and religious people cannot see eye to eye about evolution,*" Cronbach's Alpha = 0.73), and three items to measure their perceived compatibility (example item: "*People can believe in evolution and believe in God*," Cronbach's Alpha = 0.77). Students responded on a 5-point Likert scale, ranging from 1 (strongly disagree) to 5 (strongly agree). Fit index of the two-factor model and individual item loading from confirmatory factor analyses can be found in Supplemental Figure S1. To provide structural validity evidence (AERA, 2011) for the measurement of these two subscales (subscale perceived conflict between evolution and religion and subscale perceived compatibility between evolution and religion), we used confirmatory factor analysis using the R package *lavaan* (Rosseel, 2012). Fit index of the two-factor model and individual item loading from confirmatory factor analyses can be found in Supplemental Figure S1. Our model indicates a good fit based on the comparative fit index (CFI = 0.956) and root-mean-square error of approximation (RMSEA = 0.08).

Acceptance of Evolution. We hypothesized that conflict-reducing practices would increase acceptance of evolution for religious students and we used two previously developed surveys to measure different aspects of evolution acceptance.

Human Evolution Acceptance. We used the Inventory of Student Evolution Acceptance (I-SEA) to measure students' human evolution acceptance (Nadelson and Southerland, 2012) because human evolution tends to be the most contentious evolutionary context for students (Aini *et al.*, 2024; Pobiner, 2016; Grunspan *et al.*, 2021; de Lima and Long, 2023); we wanted to see whether conflict-reducing practices could have an impact on acceptance of the most contentious context of evolution. On this measure, students agree or disagree on a 5-point scale with eight items that measure acceptance of human evolution (e.g., "*like other organisms, the human species is a result of evolution from an ancestral group",* Cronbach's alpha = 0.90).

General Evolution Acceptance. To see whether conflict-reducing practices had an impact on general evolution acceptance, we used the Measure of Acceptance of the Theory of Evolution (MATE) 2.0 to measure general evolution acceptance (Cronbach's Alpha = 0.91). The MATE 2.0 includes 11 items with which students agree or disagree on a 5-point scale. While some MATE 2.0 items reference human evolution (for example, *"Modern humans have evolved from earlier nonhuman species"*), most items do not specifically reference humans and instead refer to speciation and changes in populations of organisms over time (e.g., "*The idea that new species evolve from earlier species is a scientifically valid theory.*") (Barnes *et al.*, 2022).

Understanding of Evolution. The videos that students watched contained only 10 min of instruction on the science of evolution, so we did not expect that the control or the conflict-reducing practices conditions would significantly impact student understanding of evolution. However, we wanted to measure it and include it as an outcome variable since there is a long-standing interest in the relationship between evolution understanding and evolution acceptance (Sinatra *et al.*, 2003; Dunk *et al.*, 2017; Aini *et al.*, 2024; Fiedler *et al.*, 2024). To measure the extent to which students had an accurate conceptual understanding of evolution, we used a subset of 13 items from the Evolutionary Attitudes and Literacy Survey (EALS) (Hawley *et al.*, 2011) that are meant to measure level of knowledge of evolution (e.g., *"In most populations, more offspring are born than can survive*") as well as the presence

TABLE 1. Demographic information

	n	%
Semester		
Fall 2022	1132	43.2%
Spring 2023	1491	56.8%
Religion		
Christian	1324	50%
Non-Christian religion	466	18%
Non-religious	833	32%
Agnostic	*598*	*23%*
Atheist	*235*	*9%*
Gender		
Woman	1770	67.4%
Man	788	30.1%
Nonbinary	38	1.4%
Other gender	15	0.6%
Decline to state	12	0.5%
Race/Ethnicity		
White	1423	54.3%
Latine/Latino/Latina	414	15.8%
Asian	391	14.9%
Black or African-American	160	6.1%
Other race/ethnicity	190	7.2%
Decline to state	45	1.7%
Total	**2623**	

of common misconceptions about evolution (e.g., "*Evolution is a linear progression from primitive to advanced species*"). Students were asked whether each item was true or false based on their evolution understanding. Students were also given the option to choose "I don't know" to decrease the chance that they would guess and answer correctly by chance. Students' scores were calculated by determining the number of items for which they chose the correct response (Cronbach's Alpha = 0.602, which is typically considered acceptable for a test that measures content knowledge of a domain; e.g., see Carlson *et al.*, 2010, pp. 136–138). We chose to use the EALS to measure evolution understanding because it has been used in other evolution education studies (Short and Hawley, 2015; Dunk *et al.*, 2017), has shown evidence of reliability and validity among college students (Hawley *et al.*, 2011), and the items do not appear to conflate evolution acceptance with evolution understanding.

Sample. We recruited students from 19 lower-level undergraduate biology courses in 11 different states in the United States (AL, AR, AZ, FL, GA, MI, MN, NC, NY, TN, and TX) and had a 61% response rate. Data were collected at the beginning of either the fall 2022 or spring 2023 semester. We analyzed the responses of 2623 students who 1) completed both surveys in their entirety, 2) confirmed that they watched the video in full, and 3) confirmed they gave honest responses to the survey. Half of these students identified as Christian, 18% identified with a non-Christian religion (Jewish, Muslim, Hindu, Buddhist), 23% identified as agnostic, and 9% identified as atheists. All demographic information is presented in Table 1.

Analyses

RQ 1. Do Conflict-Reducing Practices in Evolution Instruction Lead to Better Evolution Education Outcomes Compared With Evolution Instruction Without Conflict-Reducing Practices? To explore the impact of conflict-reducing practices on students' perceived conflict, perceived compatibility, evolution acceptance, and evolution understanding, we compared student outcomes from the two conditions with conflict-reducing practices to the condition with no conflict-reducing practices. Linear mixed models with random intercepts were used to test the effects of the presence and absence of conflict-reducing practices nested by course. Students' post-scores are the outcome variables controlled by pre-scores with the following model:

Model 1. Post ~ Pre + Conflict-reducing Practices (*conflict-reducing practices or no conflict-reducing practices (reference)*) + (1 | course), data = all

RQ 2. Does an Instructor Revealing a Non-Religious or Christian Identity Impact the Effectiveness of Conflict-Reducing Practices for Students of Varying Religious Identities? To explore the differential impact of conflict-reducing practices based on instructor religious identity, we compared student outcomes between two conditions with conflict-reducing practices, the one that was implemented by a Christian or non-religious instructor. We also explored how these conditions interacted with students' religious or non-religious affiliations. Again, we used random-intercept linear mixed models to test the effects of the conflict-reducing practices conditions and the interaction between students' religious affiliation and instructors' identity, nested by course. Students' post-scores are the outcome variables controlled by pre-scores, conditions, and students' religious identity with the following models:

Model 2. Post ~ Pre + Instructor religion (*Non-Religious Instructor or Christian instructor(reference)* + (1 | course), data = intervention conditions only

Model 3. Post ~ Pre + Instructor religion (*Non-Religious Instructor or Christian instructor(reference)* + Student Religion (*Atheist, Agnostic, Christian (reference), or Other Religion)* + (1|course), data = intervention conditions only

Model 4. Post ~ Pre + Instructor religion (*Non-Religious Instructor or Christian instructor(reference)* * Student Religion (*Atheist, Agnostic, Christian (reference), or Other Religion)* + (1|course), data = intervention conditions only

The rationale behind employing three distinct models lies in the complexity of the research question, aiming to fully understand the factors involved in the effectiveness of conflict-reducing practices in evolution instruction. Model 2 investigates the impact of instructor identity on post-scores, considering only the main effect of instructor religion while controlling for prescores and nesting data by course. Model 3 incorporates students' religious affiliations alongside instructor identity, recognizing the effects of both instructor and student religious identities. Finally, Model 4 investigates the interaction between instructor and student religious identities, exploring

how the combination of identities influences the effectiveness of conflict-reducing practices in evolution instruction.

The findings will be presented by addressing research questions for each outcome variable: perceived conflict between religion and evolution, perceived compatibility, acceptance of human evolution, acceptance of general evolution, and understanding of evolution. To illustrate the contrast between pre-scores and post-scores broken down by condition, we use half-violin plots. To assess the efficacy of conflict-reducing practices by instructor and student identity, we calculated estimated marginal means and did pairwise comparisons by condition to determine significant differences, for which the results are illustrated with dot and line figures. All coefficients, p values, and omnibus statistics for all regressions can be found in Supplemental Material S2.

Interpreting the Results and Limitations

There is a limitation we would like to acknowledge when interpreting results of this study. We included a measure of religiosity in our survey but decided not to incorporate this variable into the main analyses of this manuscript. Religiosity measures the extent to which students identify as religious (Cohen *et al.*, 2008) and includes four items on which students rated their agreement on a 5-point scale (e.g., "*I believe in God*" and "*I consider myself as a religious person*"). So, with this measure for instance, we could consider whether a student is Christian but also the extent to which this student identifies as religious, potentially allowing us to examine differences between practicing Christians who are likely to score high on religiosity and more cultural Christians who are likely to score low on religiosity. However, this variable is problematic when comparing with non-religious students because scores for non-religious students unsurprisingly cluster at the low end of the scale, leaving insufficient variation in this measure to explore their outcomes compared with Christian students or students from non-Christian religions. Thus, incorporating a religiosity measure into these analyses with religious affiliation would prevent us from making comparisons between students from all religious and non-religious affiliations. However, we recognize the value of this additional variable and conducted an additional analysis exploring the interaction between instructional condition and religiosity for just Christian students, for which there was one statistically significant result, and this is included in Supplemental Material S2.

Positionality of Research Team

The research team includes individuals with a variety of religious identities, including Christian, Muslim, atheist, agnostic, and spiritual. Further, the team includes women, LGBTQ+ individuals, individuals who were first generation college students, international students, and individuals who grew up as low and middle socioeconomic class. Researchers include undergraduate students (A.S. and C.E.), graduate students (R.A. and B.E) and faculty (Y.Z., S.B, and M.B), all of whom have taken biology courses and most of whom teach biology courses. The underlying commonalities that unite the research team is that we all believe that evolution is an important part of biology education, that evolution acceptance is an important goal of evolution instruction, and that there can be compatibility between religion and evolution so that one does not have to reject a belief in a higher power to accept evolution.

RESULTS

RQ1: Do conflict-reducing practices in evolution instruction lead to better student outcomes compared with evolution instruction without conflict-reducing practices?

Finding 1a: The Video Conditions That Included Conflict-Reducing Practices Reduced Students' Perceived Conflict Between Evolution and Religion and Increased Their Perceived Compatibility Compared With the Video Condition Without Conflict-Reducing Practices. After the intervention, students who were randomly assigned to receive conflict-reducing practices reported significantly lower perceived conflict between evolution and religion (Figure 2A) and significantly greater perceived compatibility compared with the control group (Figure 2B). The main effects of conflict-reducing practices were significant for reducing perceived conflict ($\beta = -0.37$, $p < 0.001$) and increasing perceived compatibility ($\beta = 0.18$, $p < 0.001$) after the intervention.

Finding 1b: The Video Conditions That Included Conflict-Reducing Practices Increased Students' Acceptance of Human Evolution Compared With the Video Condition Without Conflict-Reducing Practices. After the intervention, students who received conflict-reducing practices were significantly more likely to accept human evolution compared with the control group ($\beta = 0.04$, $p < 0.05$), while differences in general evolution acceptance between the two groups were not statistically significant ($\beta = 0.00$, $p = 0.90$) (Figure 2C).

Finding 1c: The Video Conditions That Included Conflict-Reducing Practices did not Statistically Significantly Influence Students' Evolution Understanding Compared With the Video Condition Without Conflict-Reducing Practices. The main effect of condition was not statistically significant in predicting students' evolution understanding after instruction ($\beta = 0.01$, $p = 0.06$), indicating that the evolution instructional video that included conflict-reducing practices did not increase student understanding of evolution (Figure 2E).

Research Question 2: Does an instructor revealing a non-religious or Christian identity impact the effectiveness of evolution instruction that includes conflict-reducing practices?

Finding 2: Conflict-Reducing Practices Were Equally Effective When the Instructor was Non-Religious Compared With When They Were Christian, Except in the Case of Perceived Compatibility, in which the Non-Religious Instructor More Effectively Increased Perceptions of Compatibility Among Atheist Students. When looking at the different impacts of conflict-reducing practices implemented by a non-religious or Christian instructor, two-way interactions (Model 4) between the instructor and student religious identity were not significant ($p > 0.05$) for perceived conflict, human evolution acceptance, general evolution acceptance, and evolution understanding, indicating that conflict-reducing practices were equally effective for students with different reli-

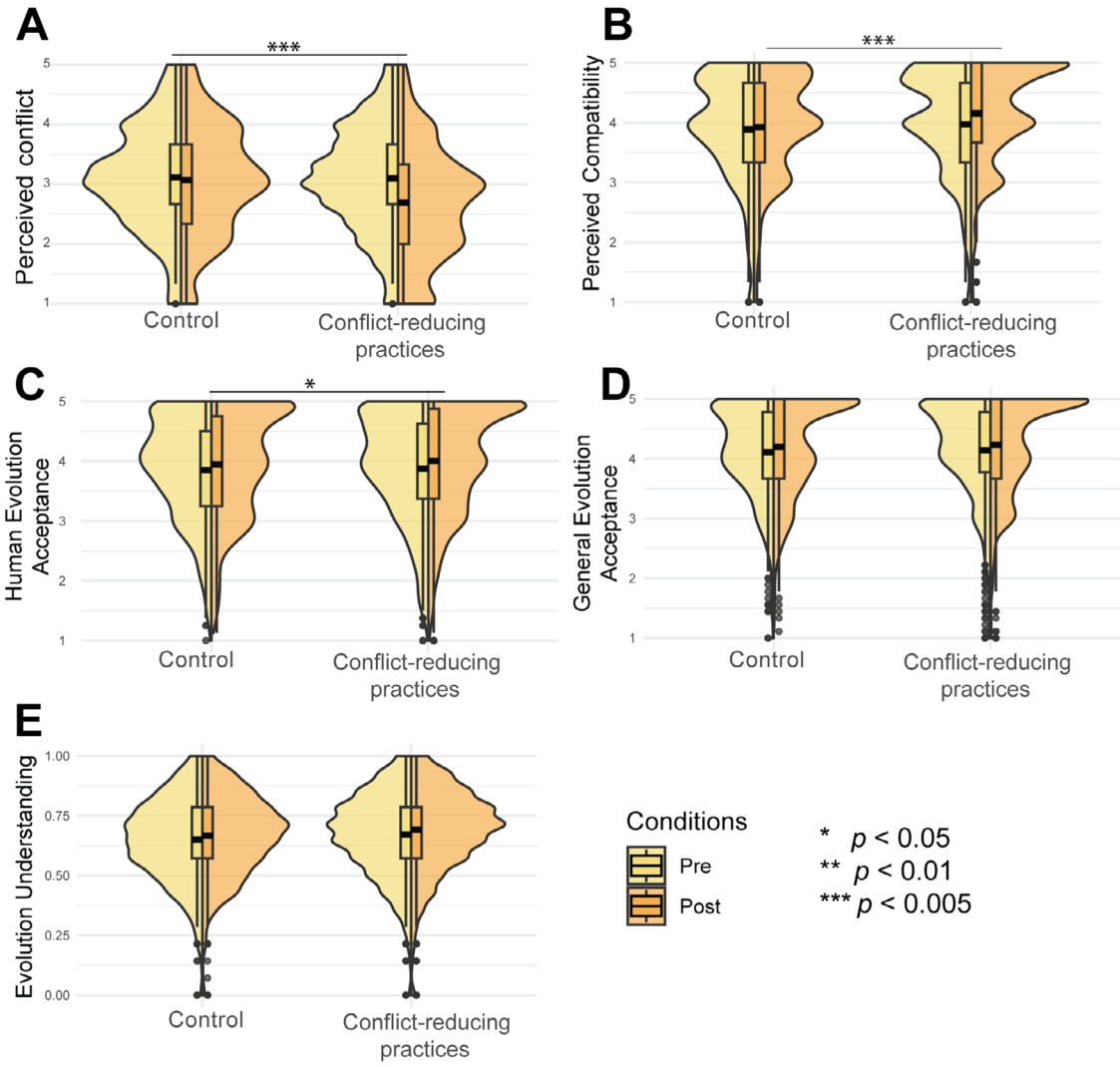


FIGURE 2. Half violin plots illustrating students' (A) perceived conflict between evolution and religion, (B) perceived compatibility between evolution and religion, (C) human evolution acceptance, (D) general evolution acceptance, and (E) evolution understanding both before and after watching either a control video that included only facts about evolution and evidence for evolution or an intervention video in which they also received conflict-reducing practices.

gious identities regardless of whether the instructor revealed themselves as Christian or non-religious. However, a significant interaction emerged for perceived compatibility between the non-religious instructor and atheist students, using the Christian instructor and Christian students as the reference group. Pairwise comparisons of estimated marginal means indicated that the non-religious instructor was more effective for increasing perceived compatibility among atheist students compared with the Christian instructor ($\beta = -0.26$, $p = 0.014$). See Figure 3 for comparisons illustrated.

DISCUSSION

Our study is the first to show through a randomized controlled design that conflict-reducing practices are effective overall for improving student perceptions of evolution by reducing conflict, increasing compatibility, and increasing human evolution acceptance. We also showed that both non-religious and Christian instructors can be effective at improving student outcomes using conflict-reducing practices with few differences between either instructor condition on these outcomes. In contrast to what might be expected based on social identity theory, non-religious instructors, who comprise the majority of evolution instructors (Ecklund and Scheitle, 2007; Pew Research Center, 2009), seem to be able to elicit positive changes among their religious students, which aligns more with intergroup contact theory. However, non-religious instructors notably have an outsized impact on atheist students' perceived compatibility between religion and evolution, which could be because of their identity alignment.

This is a relatively short intervention that takes less than 30 min outside of class time, yet we saw moderate effects on students' perceived conflict and compatibility and small effects on students' human evolution acceptance. We found these results encouraging because it indicates that biology instructors

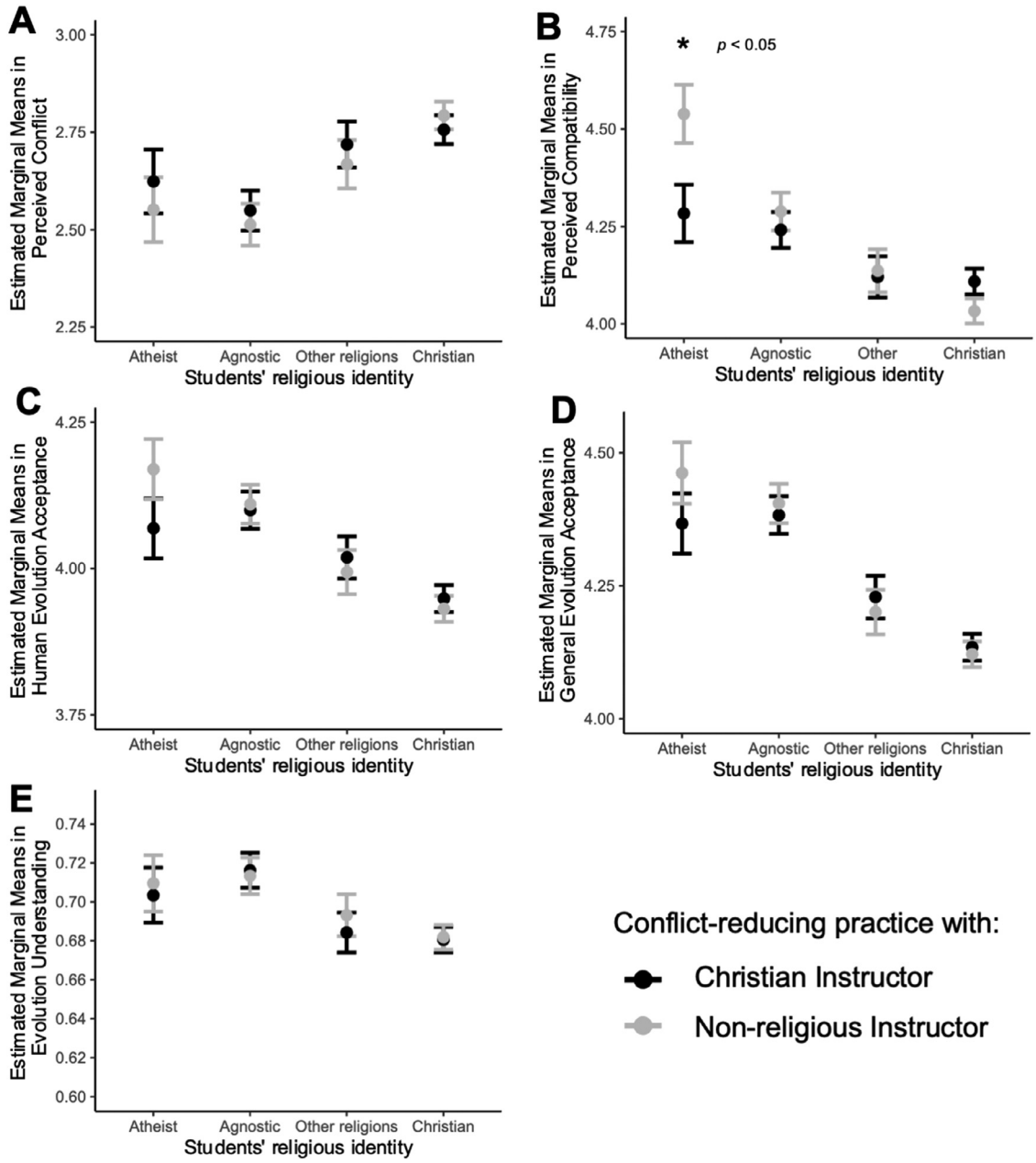


FIGURE 3. Graphs illustrating students' (A) perceived conflict between evolution and religion, (B) perceived compatibility between evolution and religion, (C) human evolution acceptance (D) general evolution acceptance, and (E) evolution understanding after watching a video that included conflict-reducing practices in evolution instruction from an instructor who revealed themselves as either Christian or non-religious. Data are disaggregated by students who identified as atheist, agnostic, Christian or other non-Christian religion (Jewish, Muslim, Hindu, Buddhist).

can have a positive impact on student evolution education outcomes with an intervention done completely outside of class. This could be important for implementing conflict-reducing practices; we have found that biology instructors can be hesitant to discuss anything about religion in the classroom because they feel as though they are not properly trained to do so or lack expertise (Barnes and Brownell, 2016). Thus, instructors themselves may not need to implement the conflict-reducing practices in the classroom and instead can use a prerecorded video of themselves or even a different instructor talking about conflict-reducing practices in the context of evolution. Further, giving students the opportunity to engage with these practices outside of class may be better for religious students to reduce the possibility that they experience negative remarks or jokes about religion or religious beliefs from their peers (Barnes and Brownell, 2016, 2018; Edwards *et al.*, 2024). We have provided the script and the slides in the Supplemental Material S3 in case instructors are

interested in creating their own videos to use outside of their classes.

We were encouraged by the short-term effects of such a small and impersonal intervention. However, future studies ought to assess whether these effects may be larger with their own instructor in the video and whether these effects persist over time to the end of their evolution instruction in the class. Further, it is likely that students will have variable evolution education experiences in undergraduate biology after receiving this intervention that could impact their perceptions of conflict, compatibility, and their acceptance of evolution, so it would be interesting to explore whether this single intervention can have a lasting impact on student evolution education outcomes or whether other experiences learning evolution without conflict-reducing practices will lessen the impact. However, the strength of the randomized controlled design is not to demonstrate long-term change, but to isolate, as close as possible, the causal mechanism of change for a variable. To minimize confounding variables, this is most possible to achieve in a short-term intervention such as this one. Future research, however, could design a study so that students receive additional surveys at additional points in a semester and over subsequent years.

We also demonstrated that non-religious instructors may have a unique ability in helping their atheist students see potential compatibility between evolution and religion, which raises several questions about why we saw this effect with this specific group and this specific variable only. One possible explanation is that non-Christian students uniquely view the non-religious instructor as more credible in science which is in line with prior research in social psychology (Rios *et al.*, 2015; Beauchamp and Rios, 2020; Mackey *et al.*, 2023). But why did we not see this same effect among agnostic students who are also not religiously affiliated or religious students who are not Christian? One explanation is that being atheist is experienced as a minority stigmatized identity outside of academic biology (Edgell *et al.*, 2006; Abbott and Mollen, 2018) which makes the identity alignment between the atheist students and non-religious instructor more salient (Tajfel and Turner, 2004). But why did we see this impact on perceived compatibility and not perceived conflict or acceptance of evolution? It may be that presenting examples of compatibility for atheist students helps them see more compatibility but does not negate their experiences of conflict between religion and evolution. Finally, atheist students' acceptance of evolution may not be differentially affected by different conditions because their acceptance of evolution is unlikely to be impacted by identity variables such as religion.

Based on social identity theory and identity alignment, we would have predicted that Christian students would have had more positive outcomes in response to the Christian instructor teaching with conflict-reducing practices compared with the non-religious instructor. However, our data do not support this because the Christian and non-religious instructor were able to equally effect positive change for Christian students. One explanation is that the Christian instructor did not elicit identity alignment with enough of the Christian students. In this study the instructor actor only revealed that they were Christian but not any other specific denomination or belief. Given the broadness and diversity of Christian cultures and belief systems, for identity alignment to occur, Christian students may need to see that the instructor shares their specific denomination or more specific religious beliefs. Particularly for students who come from denominations or cultures in which evolution rejection is high, such as among evangelical or born-again Christians, they may need to experience conflict-reducing practices from an instructor who shares this specific background. Indeed, prior work with biology students from Brigham Young University, where the majority of both the instructors and students are members of the Church of Jesus Christ of Latter-Day Saints (LDS), has shown large impacts of conflict-reducing practices on evolution acceptance (Manwaring *et al.*, 2015; Lindsay *et al.*, 2019) and that perhaps even have long term generational effects that are passed down to children of parents who received these conflict-reducing practices (Bradshaw *et al.*, 2018). Alternatively, Christian students may need to see that their instructor is also *politically* aligned to experience religious identity alignment. Given that specific sects of Christianity contain disproportionately high sentiments of homophobia/transphobia (Finlay and Walther, 2003; Pew Research Center, 2014) and outward or covert racism (Wright *et al.*, 2018; Kalinowski *et al.*, 2024), undergraduate Christian students who are social justice advocates may not align themselves with someone who identifies as Christian without knowing more information about their other beliefs. Finally, differences in religiosity levels may also explain why Christian students were not differentially impacted by the Christian instructor overall. Our supplemental analyses, which examined religiosity among Christian students, revealed that those who were highly religious were less influenced by the non-religious instructor than Christian students with lower religiosity when it came to perceived compatibility (Supplemental Material S2). This suggests that rather than experiencing a strong sense of identity alignment with the Christian instructor, highly religious Christian students may have instead experienced identity *misalignment* with the non-religious instructor. In other words, the instructor's non-religious identity may have created a perceived disconnect, making it harder for highly religious students to see them as compatible with their own worldview or values. Further, we only found this significant effect for perceived compatibility, reinforcing the idea that this variable may be particularly sensitive to identity similarities or differences between instructors and students.

Student understanding of evolution did not change in the control or the conflict-reducing practices conditions, confirming that the 10 min of facts about evolution and evidence for evolution presented in these three videos was not enough to impact how much students understood about evolution as measured by our survey. Evolution is notably a conceptually difficult topic for many students (Kampourakis and Zogza, 2009; Nehm and Ha, 2011) and the videos from this study did not incorporate active learning or deliberate practice for students, which would more likely lead to conceptual understanding gains (Haak *et al.*, 2011; Freeman *et al.*, 2014). Further, the content presented on evolution in the video was not completely aligned with the content represented in the EALS survey. For instance, the content in the video focused on the definition of evolution (change in a population overtime), the explanatory power of evolution (both diversity and unity of

life), and the multiple converging lines of evidence for evolution (fossil record, homology, biogeography, and direct observation) while the EALS scales focused on a variety of concepts such as natural selection and inheritance (see Supplemental Material for full EALS items). Although we did not expect the short video to immediately increase evolution understanding, we posit that the other improved outcomes we did see should facilitate future understanding gains. Increasing acceptance of evolution, decreasing perceived conflict between religion and evolution, and increasing perceived compatibility between religion and evolution may increase understanding of evolution in the future via an indirect effect of less hostility and increased engagement with evolution content. Whether learning about speciation in class, encountering fossils in a museum, or hearing about evolution in a nature documentary, it is likely that when students accept evolution more, perceive less conflict, and perceive more compatibility with their religious beliefs, they will be more engaged with that material. It would be valuable then to empirically document in future studies whether engagement in learning evolution indeed increases after students' acceptance of evolution increases, their perceived compatibility increases, and their perceived conflict decreases.

We found that conflict-reducing practices led to increased human evolution acceptance but not general evolution acceptance. The measure we used of general evolution acceptance was the Measure of Acceptance of the Theory of Evolution (MATE) 2.0 (Barnes *et al.*, 2022), which primarily measures macroevolutionary contexts and includes humans but is not as specific to humans on all items as our measure of human evolution acceptance was (Nadelson and Southerland, 2012). Students typically perceive the most conflict with human evolution based on their religious beliefs, so this may be why we saw differences only in human evolution acceptance but not general evolution acceptance based on the use of conflict-reducing practices. It also could be a measurement issue in that students did not adequately read the prompt that highlights human evolution in the MATE 2.0. This underscores how important the decision is for which evolution acceptance instrument researchers choose to use, which has recently been highlighted by a meeting report written by an interdisciplinary group of evolution education researchers (Barnes *et al.*, 2024). These results highlight that the choice of evolution acceptance instrument is important specifically because different instruments can yield different effects, as has been previously documented (Barnes *et al.*, 2019).

We recognize that the intervention showed promising results for perceived conflict and compatibility, modest results for human evolution acceptance, but no result for changing general evolution acceptance, which some may see as a lack of effectiveness for the intervention. We disagree: students reject human evolution at a much higher rate than other areas of evolution, perceived conflict is the strongest predictor for evolution acceptance (Barnes *et al.* 2021), and perceived conflict between religion and evolution is often highest for human evolution as students tend to not have conflict with other aspects of evolution like microevolution (Pobiner, 2016). This highlights the importance of emphasizing the intervention's impact on reducing conflict between evolution and religion, which then should primarily affect human evolution acceptance and we saw that in our results.

A strength of the randomized controlled study design is that students were randomized into different conditions, which means we can eliminate bias based on self-selection into the intervention. Additionally, this is a beneficial approach to test how changing one component (e.g., the instructor religious identity) while keeping everything else the same may affect an outcome. However, it also means that only a small number of variables can be tested in a single experiment. Given the many ways that one can implement conflict-reducing practices and what factors may influence this (e.g., instructor gender, the amount of time allotted to conflict-reducing practices, specific types of conflict-reducing practices), it was not feasible to do randomized controlled study designs to test all of them in this study. For instance, one important condition we were not able to test is having an instructor reveal as non-Christian but religious, such as a Muslim instructor. Further, we were not able to test how a condition such as this or our current conditions affected students from minority religions, such as Muslim students, due to the small sample size of these populations among college biology students in the United States. Thus, we encourage future researchers to use a variety of different approaches in concert, including quantitative pre-post quasi-experimental naturalistic study designs, qualitative interview studies, and especially randomized controlled studies, to further probe this important area of research.

CONCLUSION

In this study, we found that students' perceived conflict between evolution and religion, perceived compatibility, and human evolution acceptance were improved by using conflict-reducing practices during evolution instruction. This is the first study to our knowledge to show the efficacy of these practices in a randomized controlled design, which has allowed us to definitively isolate the impact of these practices on students' outcomes. Encouragingly, we also found that both an evolution instructor who revealed themselves as non-religious and an instructor who revealed themselves as Christian could have positive impacts on both their religious and non-religious students. Uniquely, we found that the non-religious evolution instructor video was able to increase perceived compatibility among atheist students to a greater extent than the Christian instructor video. This study underscores the utility of randomized controlled designs to illuminate how different ways of implementing conflict-reducing practices can impact the efficacy of them for students with different religious and non-religious identities.

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DATA AVAILABILITY

The R code and data supporting the conclusions of this article are available at GitHub (https://github.com/BioedSPSlab/conflict-reducing-practices-randomize).

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